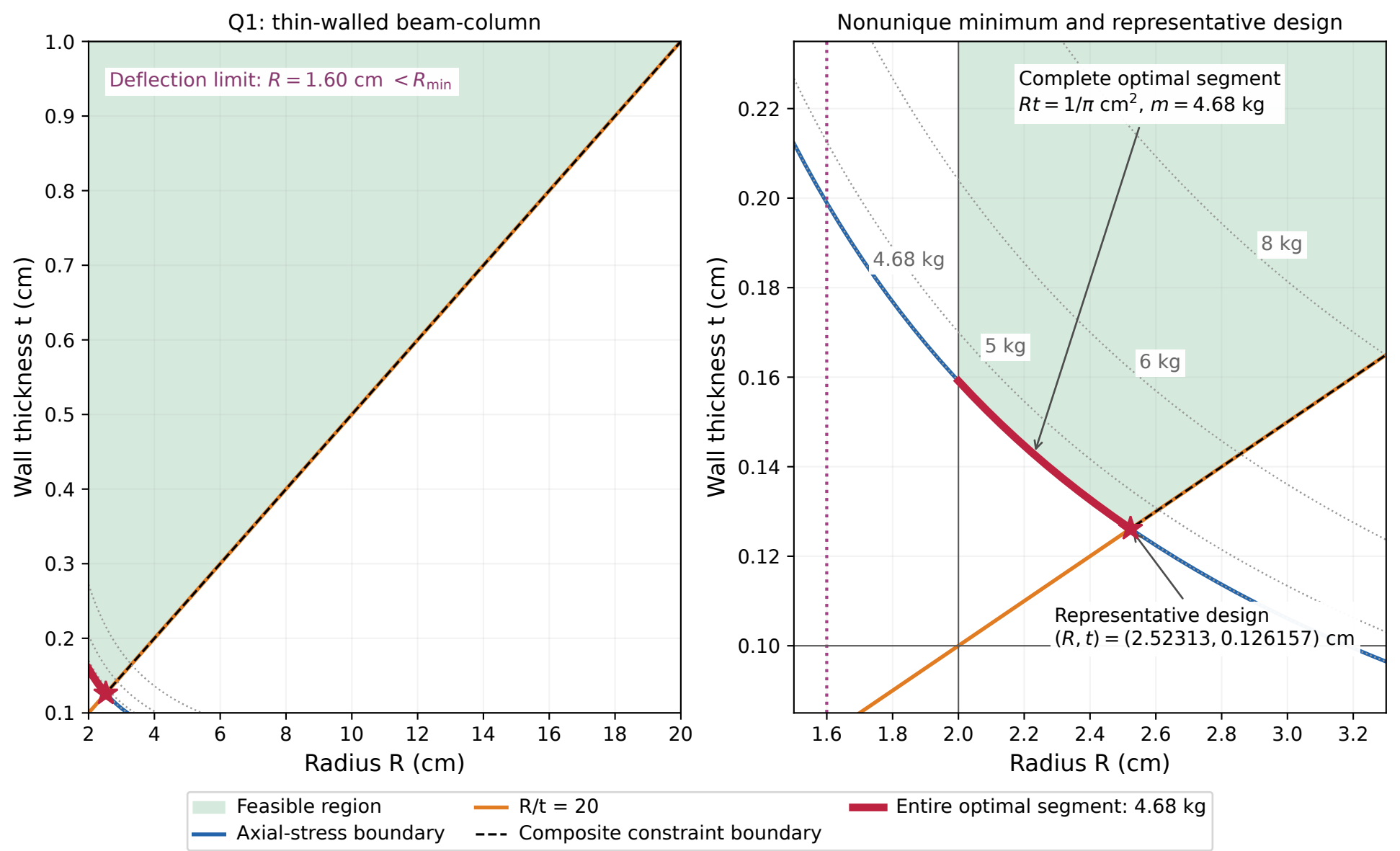

Q1: thin-walled beam-column
Deflection limit: $R = 1.60$ cm $< R_{\min}$
Wall thickness t (cm)
Radius R (cm)
Nonunique minimum and representative design
Complete optimal segment
$Rt = 1/\pi$ cm^2, $m = 4.68$ kg
4.68 kg
5 kg
6 kg
8 kg
Representative design
$(R, t) = (2.52313, 0.126157)$ cm
Wall thickness t (cm)
Radius R (cm)
Feasible region
Axial-stress boundary
R/t = 20
Composite constraint boundary
Entire optimal segment: 4.68 kg